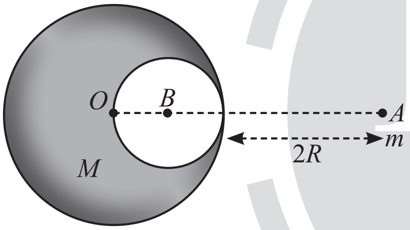
Gravitation

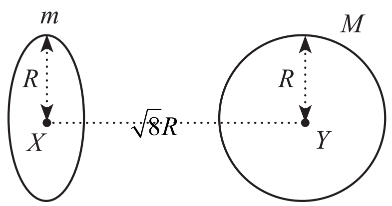
Single Correct Type Questions

1. A solid sphere of radius R gravitationally attracts a particle placed at $3R$ from its centre with a force F. Now a spherical cavity of radius $\left(\frac{R}{2}\right)$ is made in the sphere (as shown in figure) and the force becomes F_2. The value of $F_1 : F_2$ is: **[25 Feb, 2021 (Shift-I)]**



(*a*) 36:25 (*b*) 50:41 (*c*) 25:36 (*d*) 41:50

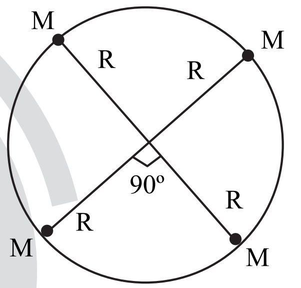
2. Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'. **[26 Feb, 2021 (Shift-I)]**



(*a*) $\frac{\sqrt{8}}{9}\times\frac{GmM}{R}$ (*b*) $\frac{1}{3\sqrt{8}}\times\frac{GMm}{R^2}$

(*c*) $\frac{\sqrt{8}}{27}\times\frac{GmM}{R^2}$ (*d*) $\frac{2\sqrt{2}}{3}\times\frac{GMm}{R^2}$

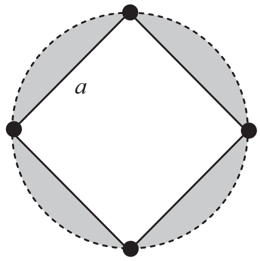
3. Four particles each of mass M, move along a circle of radius R under the action of their mutual gravitational attraction as shown in figure. The speed of each particle is: **[1 Sept, 2021 (Shift-II)]**



(*a*) $\frac{1}{2}\sqrt{\frac{GM}{R(2\sqrt{2}+1)}}$ (*b*) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}-1)}$

(*c*) $\frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}+1)}$ (*d*) $\sqrt{\frac{GM}{R}}$

4. Four identical particles of mass M are located at the corners of a square of side 'a'. What should be their speed if each of them revolves under the influence of other's gravitational field in a circular orbit circumscribing the square? **[8 April, 2019 (Shift-I)]**



(*a*) $1.21\sqrt{\frac{GM}{a}}$ (*b*) $1.41\sqrt{\frac{GM}{a}}$

(*c*) $1.16\sqrt{\frac{GM}{a}}$ (*d*) $1.35\sqrt{\frac{GM}{a}}$

5. A solid sphere of mass 'M' and radius 'a' is surrounded by a uniform concentric spherical shell of thickness $2a$ and mass $2M$. The gravitational field at distance '$3a$' from the centre will be: **[9 April, 2019 (Shift-I)]**

(*a*) $\frac{2GM}{9a^2}$ (*b*) $\frac{GM}{3a^2}$ (*c*) $\frac{GM}{9a^2}$ (*d*) $\frac{2GM}{3a^2}$

6. The radii of two planets 'A' and 'B' are 'R' and '$4R$' and their densities are ρ and $\rho/3$ respectively. The ratio of acceleration due to gravity at their surfaces ($g_A : g_B$) will be: **[11 April, 2023 (Shift-I)]**

(*a*) 1 : 16 (*b*) 3 : 16 (*c*) 3 : 4 (*d*) 4 : 3

7. T is the time period of a simple pendulum on the earth's surface. Its time period becomes xT when taken to a height R (equal to earth's radius) above the surface of earth. Then, the value of x will be: **[25 Jan, 2023 (Shift-I)]**

(*a*) 4 (*b*) 2

(*c*) $\frac{1}{2}$ (*d*) $\frac{1}{4}$

8. The acceleration due to gravity at height h above the earth if $h << R$ (radius of earth) is given by

[08 April, 2023 (Shift-II)]

(*a*) $g' = g\left(1-\frac{2h}{R}\right)$ (*b*) $g' = g\left(1-\frac{2h^2}{R^2}\right)$

(*c*) $g' = g\left(1-\frac{h}{2R}\right)$ (*d*) $g' = g\left(1-\frac{h^2}{2R^2}\right)$

9. The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be: (Given g = acceleration due to gravity at the surface of earth).

[25 June, 2022 (Shift-I)]

(*a*) $\frac{g}{2}$ (*b*) $\frac{g}{4}$ (*c*) $\frac{g}{3}$ (*d*) $\frac{g}{9}$

10. If the radius of earth shrinks by 2% while its mass remains same. The acceleration due to gravity on the earth's surface will approximately:

[28 July, 2022 (Shift-I)]

(*a*) Decrease by 2 %

(*b*) Decrease by 4%

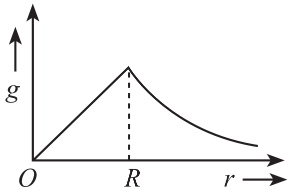
(*c*) Increase by 2%

(*d*) Increase by 4%

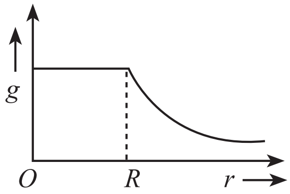
11. The variation of acceleration due to gravity (g) with distance (r) from the centre of the earth is correctly represented by: (Given R = radius of earth)

[26 June, 2022 (Shift-I)]

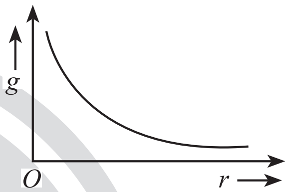
(*a*)



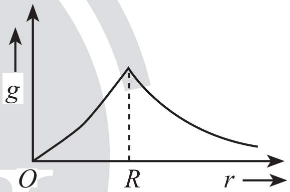
(*b*)



(*c*)



(*d*)



12. An object is taken to a height above the surface of earth at a distance 5/4 R from the centre of the earth. Where radius of earth, R = 6400 km. The percentage decrease in the weight of the object will be:

[25 July, 2022 (Shift-II)]

(*a*) 36% (*b*) 50%

(*c*) 64% (*d*) 25%

13. The acceleration due to gravity on the earth's surface at the poles is g and angular velocity of the earth about the axis passing through the pole is ω. An object is weighed at the equator and at a height h above the poles by using a spring balance. If the weights are found to be same, then h is ($h << R$, where R is the radius of the earth)

[5 Sep, 2020 (Shift-II)]

(*a*) $\frac{R^2\omega^2}{2g}$ (*b*) $\frac{R^2\omega^2}{g}$ (*c*) $\frac{R^2\omega^2}{8g}$ (*d*) $\frac{R^2\omega^2}{4g}$

14. If earth has a mass nine times and radius twice that of a planet P. Then $\frac{v_e}{3}\sqrt{x}\ ms^{-1}$ will be the minimum velocity required by a rocket to pull out of gravitational force of P, where v_e is escape velocity on earth. The value of x is

[1 Feb, 2023 (Shift-I)]

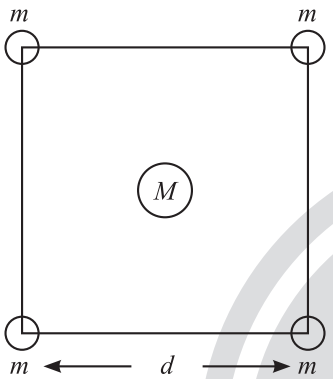
(*a*) 2 (*b*) 3 (*c*) 18 (*d*) 1

15. A body of mass m is taken from the earth surface to a height h equal to twice the radius of earth (R_e), the increase in potential energy will be: (g = acceleration due to gravity on the surface of Earth) **[25 Jan, 2023 (Shift-II)]**

(a) $3mgR_e$ (b) $\frac{1}{3}mgR_e$

(c) $\frac{2}{3}mgR_e$ (d) $\frac{1}{2}mgR_e$

16. Four spheres each of mass m form a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is: **[27 June, 2022 (Shift-II)]**



(a) $-\frac{Gm}{d}\left[(4+\sqrt{2})m+4\sqrt{2}M\right]$

(b) $-\frac{Gm}{d}\left[(4+\sqrt{2})M+4\sqrt{2}m\right]$

(c) $-\frac{Gm}{d}\left[3m^2+4\sqrt{2}M\right]$

(d) $-\frac{Gm}{d}\left[6m^2+4\sqrt{2}M\right]$

17. A body is projected vertically upwards from the surface of earth with a velocity sufficient enough to carry it to infinity. The time taken by it to reach height h is_________s. **[22 July, 2021 (Shift-II)]**

(a) $\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(b) $\frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(c) $\frac{1}{3}\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

(d) $\sqrt{\frac{R_e}{2g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2}-1\right]$

18. The masses and radii of the earth and moon are (M_1, R_1) and (M_2, R_2) respectively. Their centres are at a distance 'r' apart. Find the minimum escape velocity for a particle of mass 'm' to be projected from the middle of these two masses: **[31 Aug, 2021 (Shift-I)]**

(a) $v=\frac{1}{2}\sqrt{\frac{4G(M_1+M_2)}{r}}$ (b) $v=\sqrt{\frac{4G(M_1+M_2)}{r}}$

(c) $v=\frac{\sqrt{2G}(M_1+M_2)}{r}$ (d) $v=\frac{1}{2}\sqrt{\frac{2G(M_1+M_2)}{r}}$

19. Two planets have masses M and $16M$ and their radii are a and $2a$, respectively. The separation between the centres of the planets is $10a$. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is **[6 Sep, 2020 (Shift-II)]**

(a) $\frac{3}{2}\sqrt{\frac{5GM}{a}}$ (b) $\sqrt{\frac{GM^2}{ma}}$

(c) $2\sqrt{\frac{GM}{a}}$ (d) $4\sqrt{\frac{GM}{a}}$

20. Planet A has mass M and radius R. Planet B has half the mass and half the radius of Planet A. If the escape velocities from the Planets A and B are V_A and V_B, respectively, then $V_A/V_B = n/4$. The value of n is **[9 Jan, 2020 (Shift-II)]**

(a) 3 (b) 2

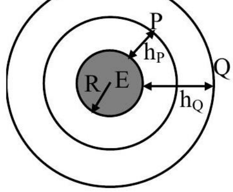
(c) 1 (d) 4

21. Two satellites of masses m and 3 m revolve around the earth in circular orbits of radii r & $3r$ respectively. The ratio of orbital speeds of the satellites respectively is: **[10 April, 2023 (Shift-I)]**

(a) 1 : 1 (b) 3 : 1

(c) $\sqrt{3}:1$ (d) 9 : 1

22. Two satellites P and Q are moving in different circular orbits around the Earth (radius R). The heights of P and Q from the Earth surface are h_P and h_Q, respectively, where $h_P = R/3$. The accelerations of P and Q due to Earth's gravity are g_P and g_Q, respectively. If $g_P/g_Q = 36/25$, what is the value of h_Q? **[JEE Adv, 2023]**



(a) $3R/5$ (b) $R/6$

(c) $6R/5$ (d) $5R/6$

23. A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R(R = radius of the earth), it ejects a rocket of mass $\frac{M}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M_e is the mass of the earth)

[7 Jan, 2020 (Shift-I)]

(*a*) $\frac{M}{20}\left(u^2-\frac{113}{200}\frac{GM_e}{R}\right)$ (*b*) $5M\left(u^2-\frac{119}{200}\frac{GM_e}{R}\right)$

(*c*) $\frac{3M}{8}\left(u+\sqrt{\frac{5GM_e}{6R}}\right)^2$ (*d*) $\frac{M}{2}\left(u-\sqrt{\frac{2GM_e}{3R}}\right)^2$

24. Two satellites, A and B, have masses m and 2m respectively. A is in a circular orbit of radius R, and B is in a circular orbit of radius $2R$ around the earth. The ratio of their kinetic energies, T_A/T_B is

[12 Jan, 2019 (Shift-II)]

(*a*) $\frac{1}{2}$ (*b*) 1

(*c*) 2 (*d*) $\sqrt{\frac{1}{2}}$

25. Two planets A and B of equal mass are having their period of revolutions T_A and T_B such that $T_A = 2T_B$. These planets are revolving in the circular orbits of radii r_A and r_B respectively. Which out of the following would be the correct relationship of their orbits?

[28 June, 2022 (Shift-I)]

(*a*) $2r_A^2 = r_B^3$

(*b*) $r_A^3 = 2r_B^3$

(*c*) $r_A^3 = 4r_B^3$

(*d*) $T_A^2 - T_B^2 = \frac{\pi^2}{GM}(r_B^3 - 4r_A^3)$

26. Consider two satellites S_1 and S_2 with periods of revolution 1 hr. and 8 hr. respectively revolving around a planet in circular orbits. The ratio of angular velocity of satellite S_1 to the angular velocity of satellites S_2 is:

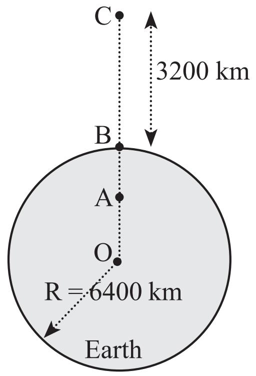
[24 Feb, 2021 (Shift-I)]

(*a*) 1:8 (*b*) 1:4 (*c*) 2:1 (*d*) 8:1

Integer Type Questions

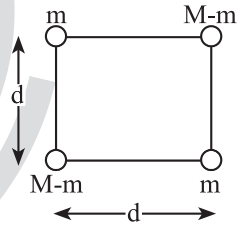
27. In the reported figure of earth, the value of acceleration due to gravity is same at point A and C but it is smaller than that of its value at point B (surface of the earth). The value of OA : AB will be x : y. The value of x is ______ .

[26 Feb, 2021 (Shift-II)]



28. A body of mass (2M) splits into four masses {m, M– m, m, M – m}, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is x : 1. The value of x is

[27 Aug, 2021 (Shift-I)]



29. The distance between two stars of masses $3M_s$ and $6M_s$ is 9 R. Here R is the mean distance between the centers of the Earth and the Sun, and M_s is the mass of the Sun. The two stars orbit around their common center of mass in circular orbits with period n T, where T is the period of Earth's revolution around the Sun. The value of n is _______.

[JEE Adv, 2021]

30. The satellites revolve around a planet in coplanar circular orbits in anticlockwise direction. Their period of revolutions are 1 hour and 8 hours respectively. The radius of the orbit of nearer satellite is 2×10^3 km. The angular speed of the farther satellite as observed from the nearer satellite at the instant when both the satellites are closest is $\frac{\pi}{x}$ rad h^{-1} where x is ___________.

[1 Sep, 2021 (Shift-II)]

ANSWER KEY

1. (*b*)	**2.** (*c*)	**3.** (*c*)	**4.** (*c*)	**5.** (*b*)	**6.** (*c*)	**7.** (*b*)	**8.** (*a*)	**9.** (*d*)	**10.** (*d*)
11. (*a*)	**12.** (*a*)	**13.** (*a*)	**14.** (*a*)	**15.** (*c*)	**16.** (*a*)	**17.** (*b*)	**18.** (*b*)	**19.** (*a*)	**20.** (*d*)
21. (*c*)	**22.** (*a*)	**23.** (*b*)	**24.** (*b*)	**25.** (*c*)	**26.** (*d*)	**27.** [4]	**28.** [2]	**29.** [9]	**30.** [3]

EXPLANATIONS

1. (*b*) $F_1 = \frac{GMm}{(3R)^2} = \frac{GMm}{9R^2}$ ...(*i*)

$$F_2 = \frac{GMm}{(3R)^2} - \frac{G\frac{M}{8}m}{\left(\frac{5R}{2}\right)^2} = \frac{GMm}{R^2}\left(\frac{1}{9} - \frac{1}{50}\right)$$

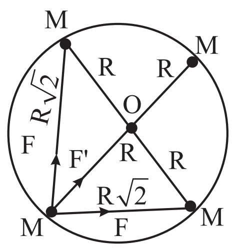
$$= \frac{GMm}{9R^2}\left(1 - \frac{9}{5}\right)$$

$$F_2 = F_1\left(1 - \frac{9}{5}\right) \text{ (Using (}i\text{))}$$

$$\frac{F_1}{F_2} = \frac{1}{1 - \frac{9}{50}} = \frac{50}{41}$$

2. (*c*) $F = M_{sphere} E_{ring} = M \times \frac{Gm \times \sqrt{8}R}{\left(R^2 + 8R^2\right)^{3/2}} = \frac{\sqrt{8}GMm}{27R^2}$

3. (*c*) $F_{net} = \frac{Mv^2}{R}$ [$\therefore$ for circular motion]



$$\sqrt{2}F + F' = \frac{Mv^2}{R}$$

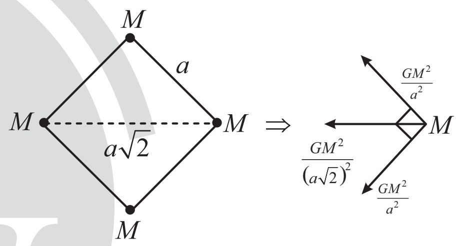
$$\sqrt{2}\frac{GMM}{(\sqrt{2}R)^2} + \frac{GMM}{(2R)^2} = \frac{Mv^2}{R}$$

$$\frac{GM}{R}\left[\frac{1}{\sqrt{2}} + \frac{1}{4}\right] = v^2$$

$$v = \sqrt{\frac{GM(4+\sqrt{2})}{R4\sqrt{2}}}$$

$$v = \frac{1}{2}\sqrt{\frac{GM}{R}(2\sqrt{2}+1)}$$

4. (*c*)



Net force on particle towards centre of circle is

$$F_C = \frac{GM^2}{2a^2} + \frac{GM^2}{a^2}\sqrt{2}$$

$$= \frac{GM^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

This force will act as centripetal force. Distance of particle from centre of circle is $\frac{a}{\sqrt{2}}$.

$$r = \frac{a}{\sqrt{2}},\ F_C = \frac{mv^2}{r}$$

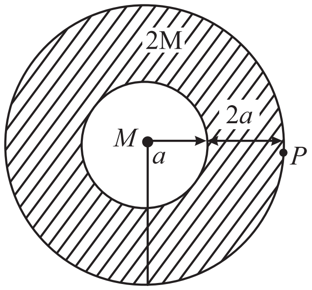
$$\frac{mv^2}{a/\sqrt{2}} = \frac{GM^2}{a^2}\left(\frac{1}{2} + \sqrt{2}\right)$$

$$v^2 = \frac{GM}{a}\left(\frac{1}{2\sqrt{2}} + 1\right)$$

$$v^2 = \frac{GM}{a}(1.35)$$

$$v = 1.16\sqrt{\frac{GM}{a}}$$

5. (*b*)



Using, Gauss's Law for gravitation
$g \cdot 4\pi r^2$ = (Mass enclosed) $4\pi G$

$$g = \frac{3M\,4\pi G}{4\pi(3a)^2} = \frac{MG}{3a^2}$$

6. (*c*) Formula of acceleration due to gravity, $g = \frac{GM}{R^2}$

$$\because M = \rho V = \rho \times \frac{4}{3}\pi R^3$$

$$g = \frac{G}{R^2} \times \rho \times \frac{4\pi}{3} R^3 = \left(\frac{4\pi}{3}G\right)\rho R$$

$$\therefore g = \frac{g_A}{g_B} = \frac{R \times \rho}{4R \times \frac{\rho}{3}} = \frac{3}{4}$$

7. (*b*) At the surface of earth time period, $T = 2\pi\sqrt{\frac{\ell}{g}}$

At a height h above the surface of earth = R

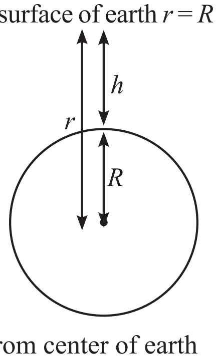
$$g' = \frac{g}{\left(1+\frac{h}{R}\right)^2} = \frac{g}{4}$$

$$\Rightarrow xT = 2\pi\sqrt{\frac{\ell}{(g/4)}}$$

$$\Rightarrow xT = 2T$$

$$\Rightarrow x = 2$$

8. (*a*) Gravity at the surface of earth $r = R + h$ and $g = \frac{GM}{R^2}$



r = distance from center of earth

$$\Rightarrow g' = \frac{GM}{r^2} = \frac{GM}{(R+h)^2} \Rightarrow g' = \frac{GM}{R^2\left(1+\frac{h}{R}\right)^2}$$

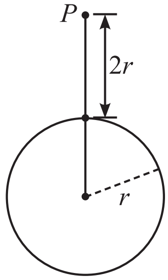
$$\Rightarrow g'(h) = \frac{GM}{R^2}\left(1+\frac{h}{R}\right)^{-2}$$

Using binominal expanssion

$$\Rightarrow g' = \frac{GM}{R^2}\left(1-\frac{2h}{R}\right)$$

$$\Rightarrow g' = g\left(1-\frac{2h}{R}\right)$$

9. (*d*)



$$g = \frac{GM}{r^2}$$

Now, $g' = \frac{GM}{(3r)^2} = \frac{g}{9}$

10. (*d*) $g = \frac{GM}{r^2}$

$$dg = -2\frac{GM}{r^3}dr$$

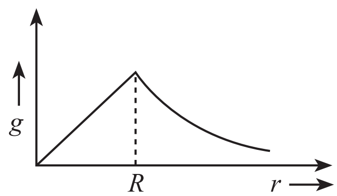
$$\Rightarrow dg = -2\left(\frac{GM}{r^2}\right)\frac{dr}{r}$$

$$\Rightarrow \frac{dg}{g} \times 100 = -2\left(\frac{dr}{r} \times 100\right) = -2 \times (-2\%) = 4\%$$

Hence, acceleration due to gravity will increase by 4%

11. (*a*) $g \propto r \quad 0 \le r \le R$

$$g \propto \frac{1}{r^2} \quad r > R$$



12. (*a*) Since, $g = \frac{GM}{R^2}$

$$g' = g\left(\frac{R}{\frac{5}{4}R}\right)^2 = \frac{16g}{25}$$

$$\% \text{ Change} = \frac{g'-g}{g}\times 100$$

$$= \frac{\frac{16}{25}g - g}{g}\times 100$$

$$= -36\%$$

13. (*a*) $g_{equator} = g - \omega^2 R$

$$g_h = g_{pole}\left(1-\frac{2h}{R}\right) = g\left(1-\frac{2h}{R}\right) = g - g\left(\frac{2h}{R}\right)$$

$\because g_h = g_{equator}$

$$g - g\left(\frac{2h}{R}\right) = g - \omega^2 R$$

$$g\left(\frac{2h}{R}\right) = \omega^2 R \quad \Rightarrow h = \frac{\omega^2 R^2}{2g}$$

14. (*a*) Escape velocity of planet $v_{\text{(escape) plant}} = \sqrt{\frac{2GM_P}{R_P}}$

$$= \sqrt{\frac{2G\left(\frac{M_e}{9}\right)}{\left(\frac{R_e}{2}\right)}}$$

$$= \frac{v_e\sqrt{2}}{3}$$

$\Rightarrow \quad x = 2$

15. (*c*) $U = \frac{-GM_e m}{r}$

Initial gravitational potential energy of the object

$$U_i = \frac{-GM_e m}{R_e}$$

Final gravitational potential energy of the object

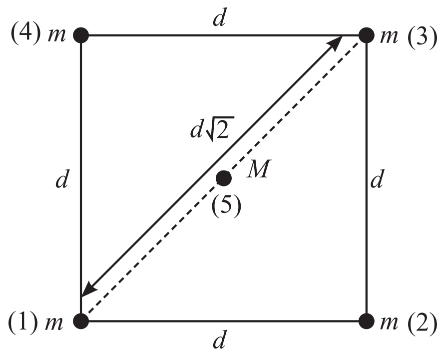
$$U_f = \frac{-GM_e m}{(R_e+h)} = \frac{-GM_e m}{R_e+2R_e} = \frac{-GM_e m}{3R_e}$$

Increase in internal energy $\Delta U = U_f - U_i$

$$= \frac{2}{3}\frac{GM_e m}{R_e}$$

$$= \frac{2}{3}mgR_e$$

16. (*a*) Amount of work done to bring mass from infinity to point 1, $U_1 = 0$



Similarly, to bring mass 2, $U_2 = \frac{-Gm}{d}m$

To bring mass 3

$$U_3 = \frac{-Gmm}{d} - \frac{Gmm}{d\sqrt{2}} = -\frac{Gm}{d}\left[\frac{m}{1}+\frac{m}{\sqrt{2}}\right]$$

To bring mass 4

$$U_4 = \frac{-Gmm}{d} - \frac{Gmm}{d\sqrt{2}} - \frac{Gmm}{d} = \frac{-Gm}{d}\left[2m+\frac{m}{\sqrt{2}}\right]$$

To bring mass 5

$$U_5 = -4\times\frac{GmM}{\left(\frac{d}{\sqrt{2}}\right)} = \frac{-Gm}{d}\left[4\sqrt{2}M\right]$$

$$U_T = U_1 + U_2 + U_3 + U_4 + U_5$$

$$U_T = \frac{-Gm}{d}\left[\left(4+\sqrt{2}\right)m + 4\sqrt{2}M\right]$$

17. (*b*) At the surface, escape velocity = $\sqrt{\frac{2GM}{R_e}}$

Applying conservation of energy:

$$\Rightarrow \frac{-GMm}{R_e} + \frac{1}{2}m\left(\sqrt{\frac{2GM}{R_e}}\right)^2 = \frac{-GMm}{r} + \frac{1}{2}mv^2$$

$$\Rightarrow \frac{1}{2}mv^2 = \frac{GMm}{r} \Rightarrow v^2 = \frac{2GM}{r}$$

$$\Rightarrow v = \sqrt{\frac{2GM}{r}}$$

$$\Rightarrow \frac{dr}{dt} = \sqrt{\frac{2GM}{r}}$$

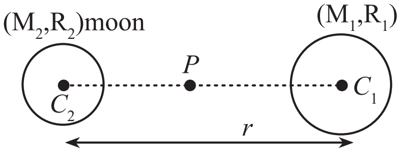
Time t taken by body to reach at hight h.

$$\therefore \int_{R_e}^{R_e+h} r^{1/2}dr = \sqrt{2GM}\int_0^t dt$$

$$\Rightarrow \frac{2}{3}\left[(R_e+h)^{3/2} - R_e^{\,3/2}\right] = \sqrt{2GM}\,t$$

$$\Rightarrow t = \frac{1}{3}\sqrt{\frac{2R_e}{g}}\left[\left(1+\frac{h}{R_e}\right)^{3/2} - 1\right]$$

18. (*b*) We have given that the masses and radii of the earth and moon are $(M_1\ R_1)$ and (M_2, R_2)



$$KE_P + PE_P = \frac{1}{2}mv_P^2 + \left\{\frac{-GM_1m}{(r/2)} - \frac{GM_2m}{r/2}\right\}$$

At infinity

$KE_\infty + PE_\infty = 0 + 0$

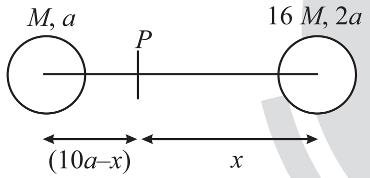
So, according to conservation of mechanical energy

$KE_P + PE_P = KE_\infty + PE_\infty$

$$\frac{1}{2}mv_p^2 - \frac{2Gm}{r}(M_1 + M_2) = 0 + 0$$

$$\Rightarrow v_P = \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

19. (*a*)



Velocity should be given so as to reach a point where field is zero.

$$\frac{16M}{x^2} = \frac{M}{(10a - x)^2}$$

$x = 8a$

By *CoE*

$$-\frac{GMm}{8a} - \frac{16GMm}{2a} + \frac{1}{2}mv^2 = -\frac{16GMm}{8a} - \frac{GMm}{2a} + 0$$

$$\frac{1}{2}mv^2 = \frac{45GMm}{8a}$$

$$v = \frac{3}{2}\sqrt{\frac{5GM}{a}}$$

20. (*d*) $v_e = \sqrt{\dfrac{2GM}{R}}$

$$\therefore \frac{v_1}{v_2} = \frac{\sqrt{\dfrac{2GM}{R}}}{\sqrt{\dfrac{2GM/2}{R/2}}} = 1 = \frac{n}{4}$$

$\Rightarrow n = 4$

21. (*c*) Orbital speed, $v = \sqrt{\dfrac{GM}{r}}$ (M : Mass of Earth)

$\Rightarrow$ Orbital speed is independent of mass of the satellite.

$$v \propto \frac{1}{\sqrt{r}}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{r_2}{r_1}} = \sqrt{\frac{3r}{r}}$$

22. (*a*) Given, $\dfrac{g_P}{g_Q} = \dfrac{36}{5}$

$$\frac{g_P}{g_Q} = \frac{\dfrac{GM}{r_P^2}}{\dfrac{GM}{r_Q^2}} = \left(\frac{r_Q}{r_p}\right)^2$$

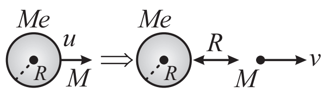
$$\frac{36}{25} = \left(\frac{r_Q}{r_P}\right)^2 \Rightarrow \frac{r_Q}{r_P} = \frac{6}{5}$$

$$r_Q = \frac{6}{5}r_P$$

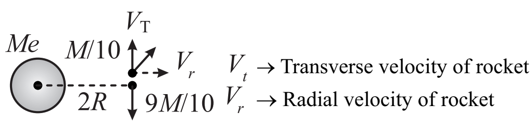
$$R + h_Q = \frac{6}{5}\left(R + \frac{R}{3}\right)$$

$$h_Q = \frac{24}{15}R - R = \frac{9}{15}R = \frac{3}{5}R$$

23. (*b*) $\dfrac{-GMeM}{R} + \dfrac{1}{2}Mu^2 = \dfrac{-GMeM}{2R} + \dfrac{1}{2}Mv^2$



$$v = \sqrt{u^2 - \frac{GM_e}{R}}$$



$$\frac{M}{10}V_T = \frac{9M}{10}\sqrt{\frac{GM_e}{2R}}$$

$$\frac{M}{10}V_r = M\sqrt{u^2 - \frac{GM_e}{R}}$$

Kinetic energy

$$= \frac{1}{2}\frac{M}{10}\left(V_T^2 + V_r^2\right) = \frac{M}{20}\left(81\frac{GM_e}{2R} + 100u^2 - 100\frac{GM_e}{R}\right)$$

$$= \frac{M}{20}\left(100u^2 - \frac{119GM_e}{2R}\right)$$

$$= 5M\left(u^2 - \frac{119GM_e}{200R}\right)$$

24. (*b*) $KE_A = \frac{1}{2}m\left(\frac{GM}{R}\right)$

$$KE_B = \frac{1}{2}(2m)\left(\frac{GM}{2R}\right)$$

$$\Rightarrow \frac{KE_A}{KE_B} = 1$$

25. (*c*) Since, $T^2 \propto R^3$

$$\Rightarrow \frac{T_A^2}{T_B^2} = \frac{r_A^3}{r_B^3} \text{ or } r_A^3 = 4r_B^3$$

26. (*d*) We know that, $\omega = \frac{2\pi}{T} \Rightarrow \frac{\omega_1}{\omega_2} = \frac{T_2}{T_1} = 8$

27. [4] Acceleration due to gravity at $A = \frac{GM}{\left(\frac{3R}{2}\right)^2}$

Acceleration due to gravity at $B = \frac{GM}{R^3} \times r$

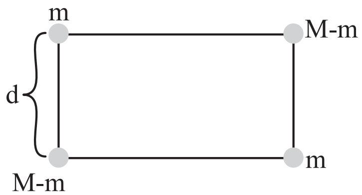
$$\because \frac{GM}{\left(\frac{3R}{2}\right)^2} = \frac{GM}{R^3} \times r$$

$$\therefore r = \frac{4R}{9} = OA$$

$$\therefore AB = R - \frac{4R}{9} = \frac{5R}{9}$$

$$\Rightarrow OA : AB = 4 : 5 = x : y \Rightarrow x = 4$$

28. [2]



$$U = -\frac{Gm(M-m)}{d} \times 4 + -2 \times \frac{Gm^2}{(d\sqrt{2})^2}$$

For U to be maximum,

$$\frac{dU}{dm} = 0 \Rightarrow m = \frac{M}{2} \Rightarrow \frac{M}{m} = 2$$

29. [9] Circular orbits

$$T = 2\pi\sqrt{\frac{R^3}{GM_S}}$$

Binary stars

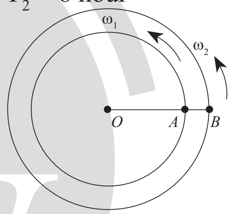
$$nT = 2\pi\sqrt{\frac{(9R)^3}{G(3M_S + 6M_S)}}$$

$$n \times 2\pi\sqrt{\frac{R^3}{GM_S}} = 9 \times 2\pi\sqrt{\frac{R^3}{GM_S}}$$

30. [3] $T_1 = 1$ hour [For A satellite]

So $\omega_1 = 2\pi$ red/hour

$T_2 = 8$ hour [For B satellite]



$\omega_2 = \frac{\pi}{4}$ rad/hour

As $T^2 \alpha R^3$

$$\left(\frac{R_1}{R_2}\right)^3 = \left(\frac{T_1}{T_2}\right)^2 = \left(\frac{1}{8}\right)^2$$

$$\frac{R_1}{R_2} = \left(\frac{1}{8}\right)^{2/3} = \left(\frac{1}{2}\right)^2$$

$\Rightarrow R_2 = 8 \times 10^3$ Km

$V_1 = \omega_1 R_1 = 4\pi \times 10^3$ Km/h

$V_2 = \omega_2 R_2 = 2\pi \times 10^3$ Km/h

Relative $\omega = \frac{V_1 - V_2}{R_2 - R_1} = \frac{2\pi \times 10^3}{6 \times 10^3}$

$\Rightarrow \frac{\pi}{3}$ rad/hour

$x = 3$